

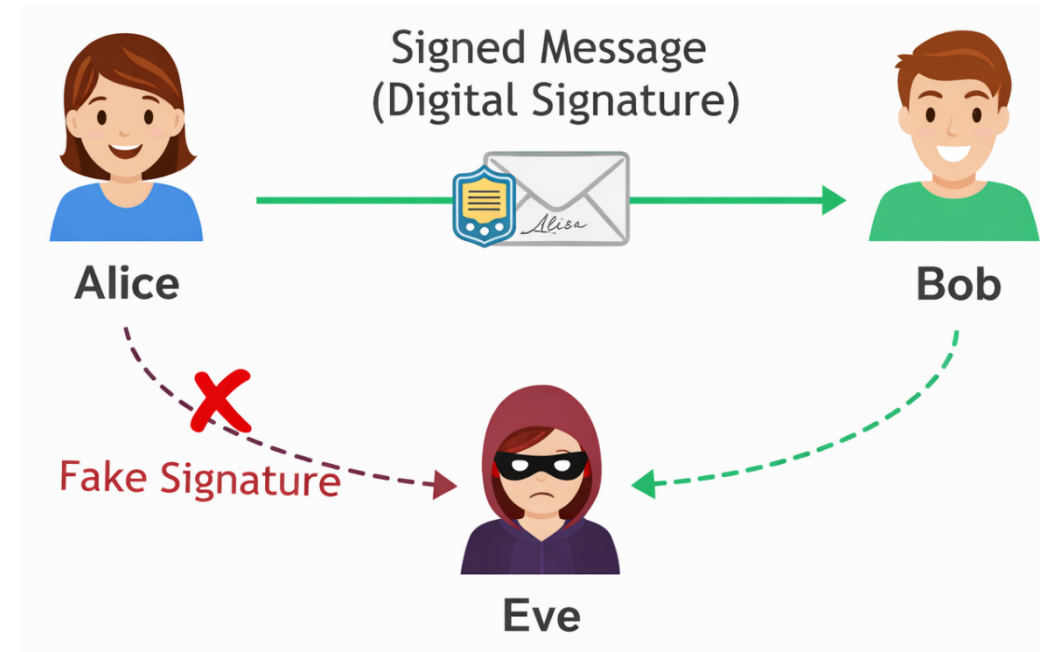


Digital Signature



Digital Signature

- A Digital Signature is an electronic method used to verify the **identity of the sender**.
- When Alice sends a message to Bob, Bob must ensure that the message **really came from Alice and not from an attacker like Eve**.
- Alice signs the message electronically using a **Digital Signature**.
- Bob verifies the signature to confirm that:
 - The message is **from Alice**
 - The message **was not modified** during transmission.
- Therefore, a Digital Signature provides **authentication and message integrity**.



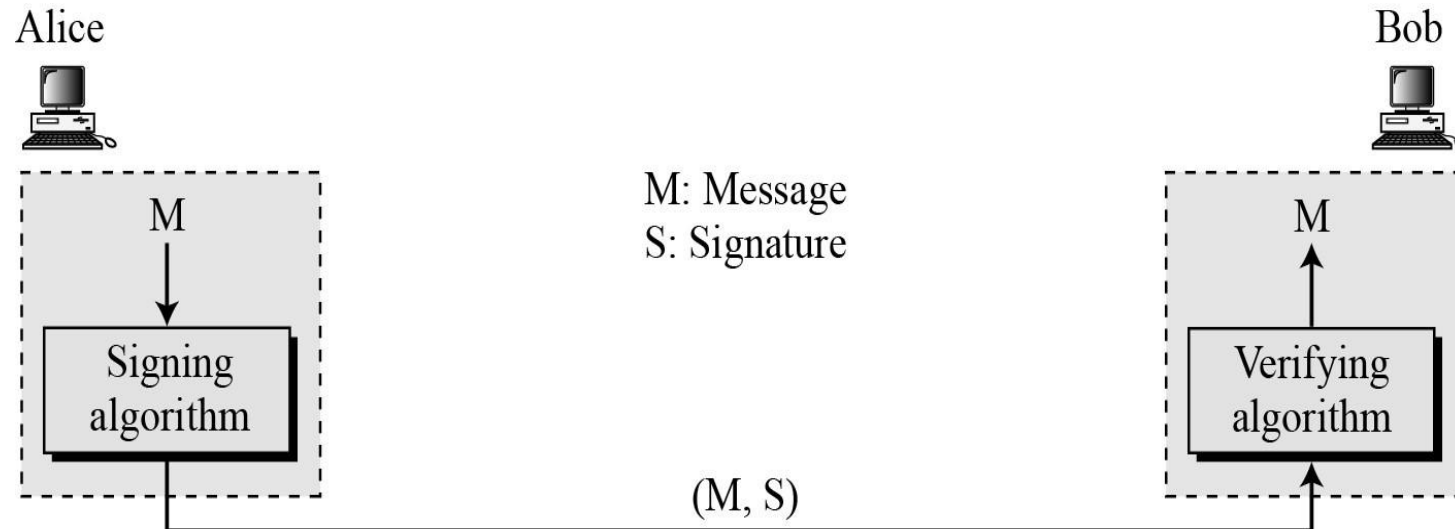


Comparison

Feature	Conventional Signature	Digital Signature
Form	Written by hand on paper documents	Electronic signature used in digital messages
Attachment	Signature is part of the document	Signature is attached electronically with the message
Verification	Verified by comparing with stored sample signature	Verified using cryptographic algorithms
Relationship	One signature can be used for many documents	Each message has a unique digital signature
Copy / Duplicate	Original and copy can be distinguished	Original and copy look the same unless verified

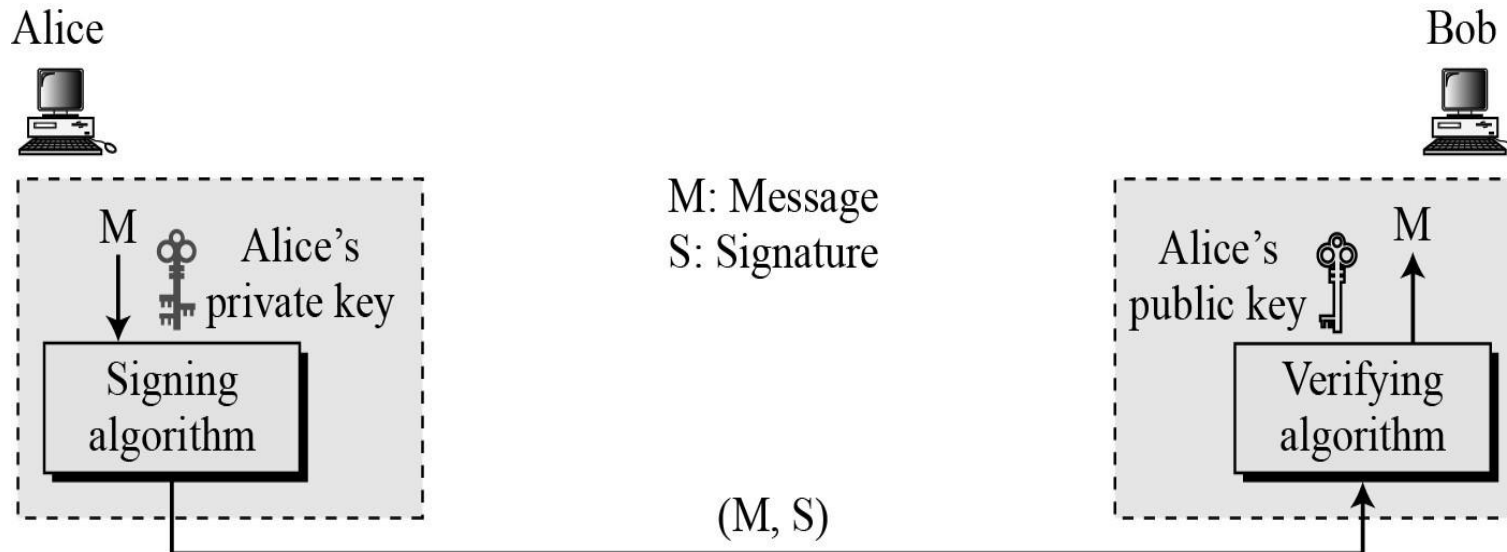
Digital Signature Process

- The sender creates a **message (M)**.
- The sender uses a **signing algorithm** to generate a **digital signature (S)**.
- The **message and signature (M, S)** are sent to the receiver.
- The receiver applies a **verification algorithm** to check the signature.
- If the signature is **valid**, the message is accepted.
- If the signature is **invalid**, the message is rejected.



Need for Keys in Digital Signature

- Digital signatures use **two types of keys: Private Key and Public Key**.
- The **sender uses the private key** with a signing algorithm to create the digital signature.
- The **receiver uses the sender's public key** with a verifying algorithm to check the signature.
- The **private key must be kept secret** by the sender.
- The sender should **not use the public key for signing**, otherwise anyone could forge the signature.





Need for Keys in Digital Signature

- A **single secret (symmetric) key** cannot be used for signing and verification.
- Secret keys are **shared between only two users**, which makes key management difficult.
- If both users share the same key, **one user can pretend to be the other**.
- Therefore, digital signatures use a **public-key system**.

Important Points

- The **sender signs the message using a private key**.
 - The **receiver verifies the signature using the sender's public key**.
- 



Signing the Digest

- Public Key Cryptography is **slow** for **large messages**.
- Instead of signing the whole message, we first create a **message digest using a hash function**.
- A **message digest is a short fixed-length value that uniquely represents the message**.
- The sender **signs the digest using their private key**.
- The receiver **creates a digest of the received message and verifies the signature using the sender's public key**.
- If both digests match, the message is **authentic and not modified**.

Advantages

- Faster processing
- Less computation
- Maintains strong security
- Detects message tampering

Signing the Digest

Sender Process (Alice)

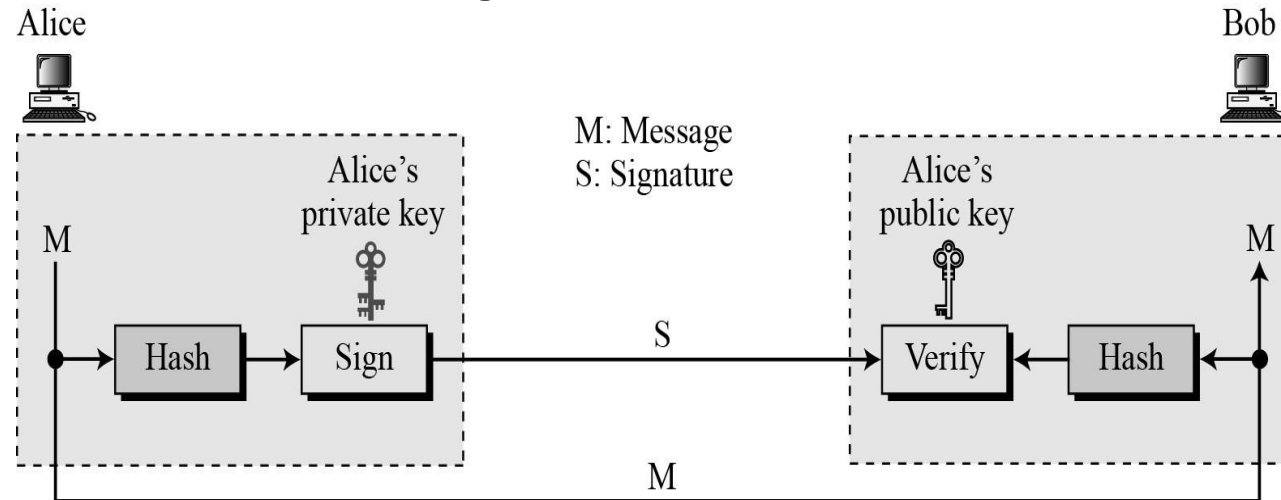
- Alice creates a **message (M)**.
- She generates a **digest** using a **hash function**.
- She **signs the digest** using her **private key**.
- She sends **Message + Signature** to Bob.

Important:

Only the **digest is signed**, not the whole message.

Receiver Process (Bob)

- Bob receives **Message + Signature**.
- He creates a **digest of the received message**.
- He uses **Alice's public key** to **verify the signature**.
- If both digests match → the **message is authentic**.





Services

A digital signature provides important security services for communication.

1. Message Authentication

- Confirms the **identity of the sender**.
- The receiver verifies the signature using the **sender's public key**.

2. Message Integrity

- Ensures the **message is not modified during transmission**.
- Any change in the message will cause **signature verification to fail**.

3. Non-Repudiation

- The sender **cannot deny sending the message** later on.
- Only the sender's **private key** can create the signature.

Note:

Digital signatures provide authentication and integrity, but **confidentiality requires encryption**.

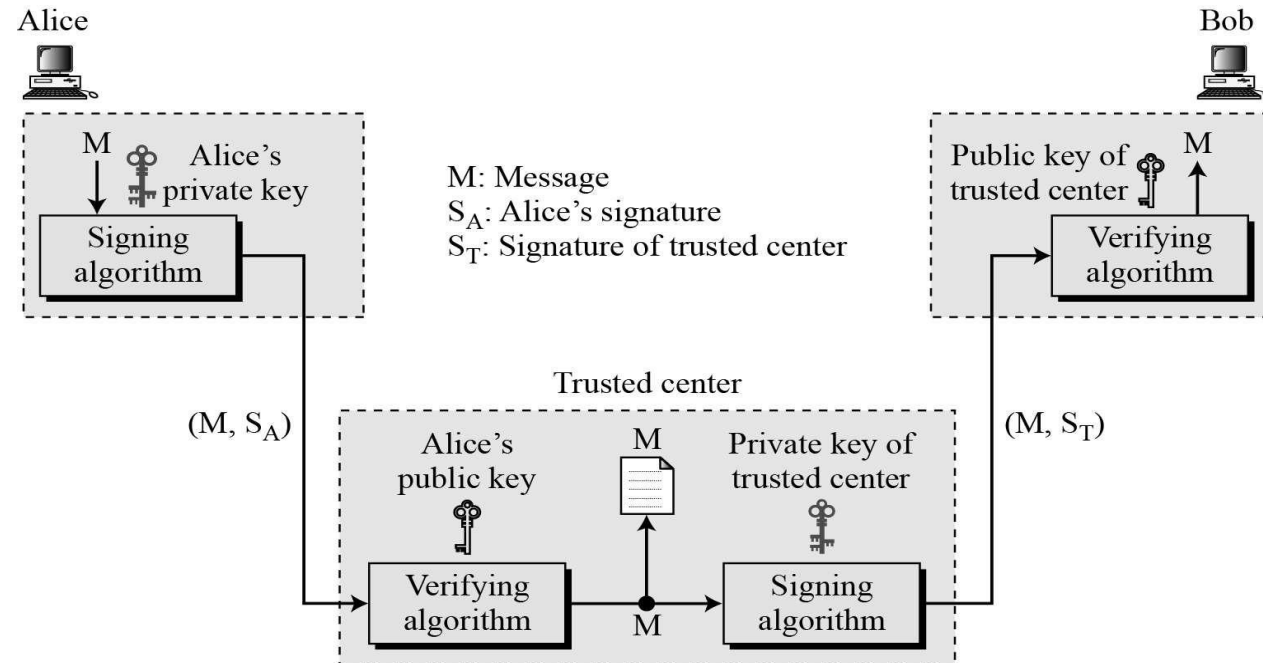
Services

Non-Repudiation Example:

- Alice sends a signed message to a bank requesting a money transfer.
- Later she cannot deny sending the message because the **signature proves her identity**.

Solution for strong proof

- Use a **Trusted Third Party (Trusted Center)** to verify and record signatures.



Services

4. Confidentiality

Digital signatures **do not provide privacy**.

- The message can still be read by others.

To achieve confidentiality:

1. Alice **signs the message using her private key**.
2. She **encrypts the message using Bob's public key**.
3. Bob **decrypts using his private key**.
4. Bob **verifies the signature using Alice's public key**.

Result:

- Message is **secure, authentic, and confidential**.

